Probability and Statistics
Practice midterm exam 2: Random variables

Date: 22 March 2017

Name: _____ **Group:** _____

Problem 1 (5 pt). MULTIPLE CHOICE questions:

- If X and Y are independent **r.v.** with $\mathsf{Var}(X) = \mathsf{Var}(Y) = 1$, then $\mathsf{Var}(2X - Y)$ is equal to

  (a) 1 (b) 3 (c) 5 (d) ..... your variant

- If $X \sim \mathscr{N}(0,1)$, then $Y = aX + b \sim \mathscr{N}(\mu, \sigma^2)$ with

  (a) $\mu = a, \sigma^2 = b$ (b) $\mu = b, \sigma^2 = a$ (c) $\mu = b, \sigma^2 = a^2$ (d) ..... your variant

- The formula $f(x) = ce^{-2x^2}$ (with a suitable constant c) gives the probability density function for the distribution

  (a) $\mathscr{E}(2)$ (b) $\mathscr{E}(\sqrt{2})$ (c) $\mathscr{N}(0, \frac{1}{2})$ (d) ..... your variant

- If $\mathsf{E}(X) = \mathsf{E}(Y)$, then $\mathsf{E}(X^2) = \mathsf{E}(Y^2)$

  (choose True or False) | T | F |

- If X and Y are independent and have distribution $\mathscr{U}(0,1)$, then $X_1 + X_2$ is uniformly distributed on the interval $(0, 2)$

  (choose True or False) | T | F |

Problem 2 (5 pt). Complete the following definitions/statements:

- A **r.v.** X is continuous if .....
  .....

- **R.v.**'s X and Y are independent if .....
  .....

- *Expectation* of a continuous **r.v.** X is equal to .....
  .....

- The probability density function of distribution $\mathscr{E}(2)$ is equal to.....
  .....

- The cumulative distribution function of a **r.v.** X is given by .....
  .....

1	2	3	4	5	6	Σ

Problem 3 (10pt)**.** Assume that Z has the standard normal distribution $\mathscr{N}(0,1)$ and $Y = 3 - 2Z$

(a) Determine the cumulative distribution function of Y in terms of Φ.

(b) Find the probability density function of Y and identify the distribution of Y.

(c) Express in terms of Φ the probability that $Y \leq 1$.

(d) Express in terms of Φ the probability that $Y > 5$.

Problem 4 (10pt)**.** Random variables X_1 and X_2 are independent and have Poisson distribution $\mathscr{P}(\lambda)$.

(a) Find $\mathsf{E}(X_1 - X_2)$ and $\mathsf{Var}(X_1 - X_2)$.

(b) Find the probability generating function of X_1.

(c) Using (b), prove that $Y := X_1 + X_2$ has distribution $\mathscr{P}(2\lambda)$.

(d) Prove that the conditional distribution $(X_1 \mid Y = n)$ of X_1 given that $Y = n$ is binomial $\mathscr{B}(\frac{1}{2})$.

Problem 5 (10pt)**.** There are two types of batteries in a bin. When in use, type i batteries last (in hours) an exponentially distributed time with rate λ_i, $i = 1, 2$. A battery that is randomly chosen from the bin will be a type i battery with probability p_i, $p_1 + p_2 = 1$.

(a) What is the probability that a randomly chosen battery is still operating after t hours of use?

(b) Calculate the expected value of operating time for a randomly chosen battery.

(c) If a randomly chosen battery is still operating after t hours of use, what is the probability that it will still be operating after an additional s hours?

Problem 6 (10pt)**.** The joint probability density function of random variables X and Y is given by the formula

$$f_{X,Y}(x,y) = \begin{cases} ax, & 0 \leq y \leq x \leq 1; \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a.

(b) Calculate the marginal probability density functions f_X and f_Y and find $\mathsf{E}(X)$ and $\mathsf{E}(Y)$

(c) Calculate the probability density function $f_{X|Y}(x, 0.5)$ of X given that $Y = 0.5$

(d) Find the conditional expectation $\mathsf{E}(X \mid Y = 0.5)$.